

Exploring the relationship between efflux and swarming in uropathogenic *E. coli*

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Introduction

The bacterial movements swimming and swarming are powered by flagella and the proton motive force. Swimming is an individual bacterial movement through liquid, and swarming is a coordinated movement on a solid surface. Various uropathogenic microbes are capable of swarming, a virulence factor in catheter-associated urinary tract infections. The uropathogenic strain *Escherichia coli* CFT073 is capable of both swimming and swarming. The CFT073 $\Delta toI/C$ mutant can swim but loses swarming capability. TolC is an efflux pump in the outer membrane of bacterial cells, and the mechanism of its involvement in swarming motility is unknown. We hypothesize that TolC is required to secrete a substance that helps in swarming such as a surfactant or quorum sensing molecule.

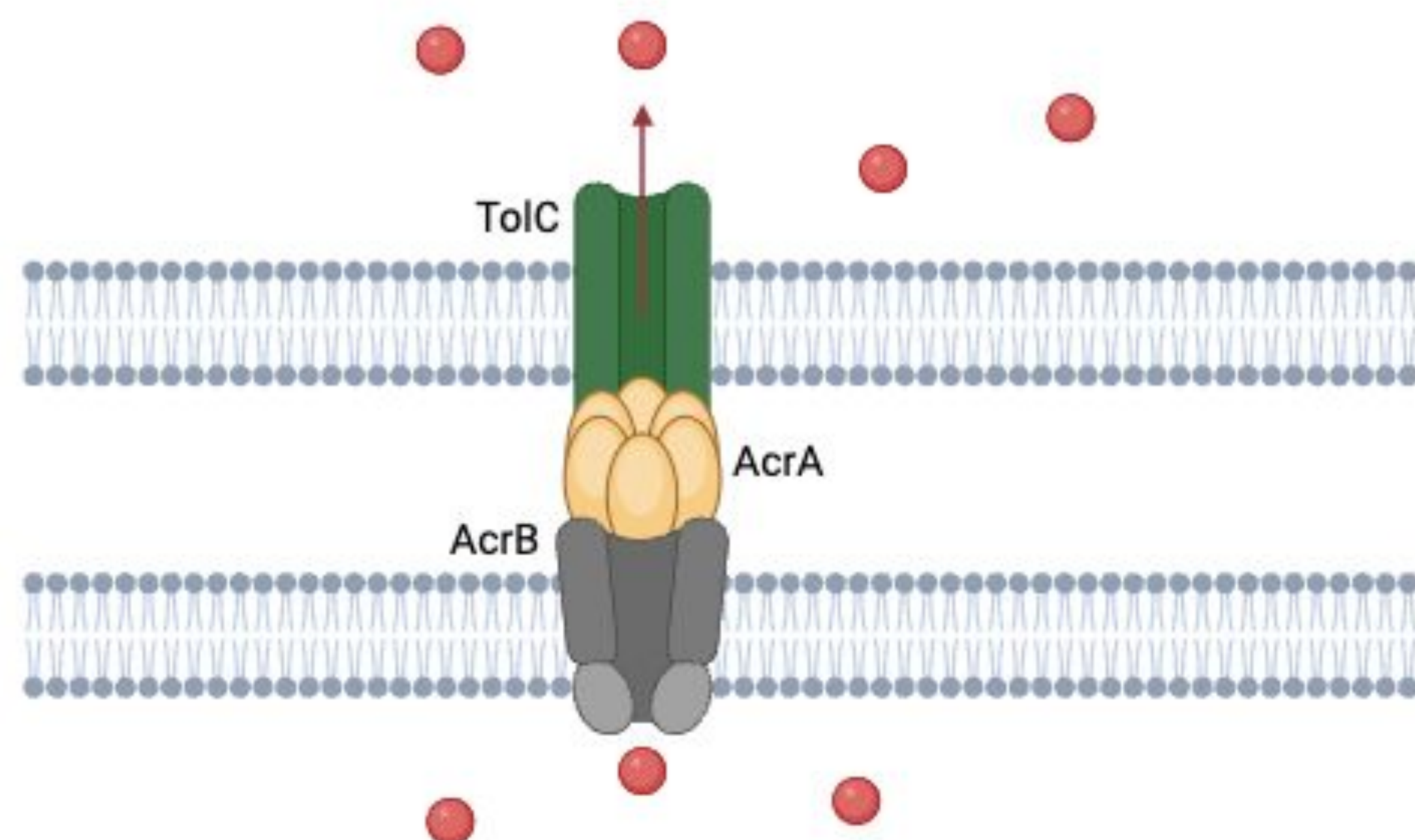


Figure I. The AcrAB-TolC efflux pump is embedded in the cell membrane of gram negative bacteria, facilitating in the export of molecules out of the cell. The TolC protein is located in the outer membrane of *E. coli*. This image was created with Biorender.com.

Methods

Overall Procedure:

- Collected wild type *E. coli* CFT073 $\Delta toI/C$ from growth plates for inoculation in TSB media and incubated at 37 deg
- Poured swarming agar plates and left covered at room temperature overnight
- Took the TSB cultures to an OD of 1
- Spun down samples and filtered/exchanged supernatants
- Inoculated swarming plates and incubated overnight at 37 degrees
- Took images of these plates on day 3 and analyzed using ImageJ

Supernatant from Solid Media:

- Wild type strains were grown on solid Eiken media
- Lifted cells from the plates the following day using water
- Spun and filtered for the supernatant switch

Surfactants:

- Added differing concentrations of Triton X to the media before plates were poured
- For these trials, no swapping of supernatants was performed

Acknowledgements

Thank you to the William and Linda Frost Fund for its continued support of BCSM undergraduate research and to the Center for Applications in Biotechnology for providing us the facilities to carry out this research.

Results

First, we established the motility of each strain on Eiken agar media, a semi-solid surface that elicits swarming movement patterns. These controls support previous knowledge that the $\Delta toI/C$ is unable to swarm (Fig II).

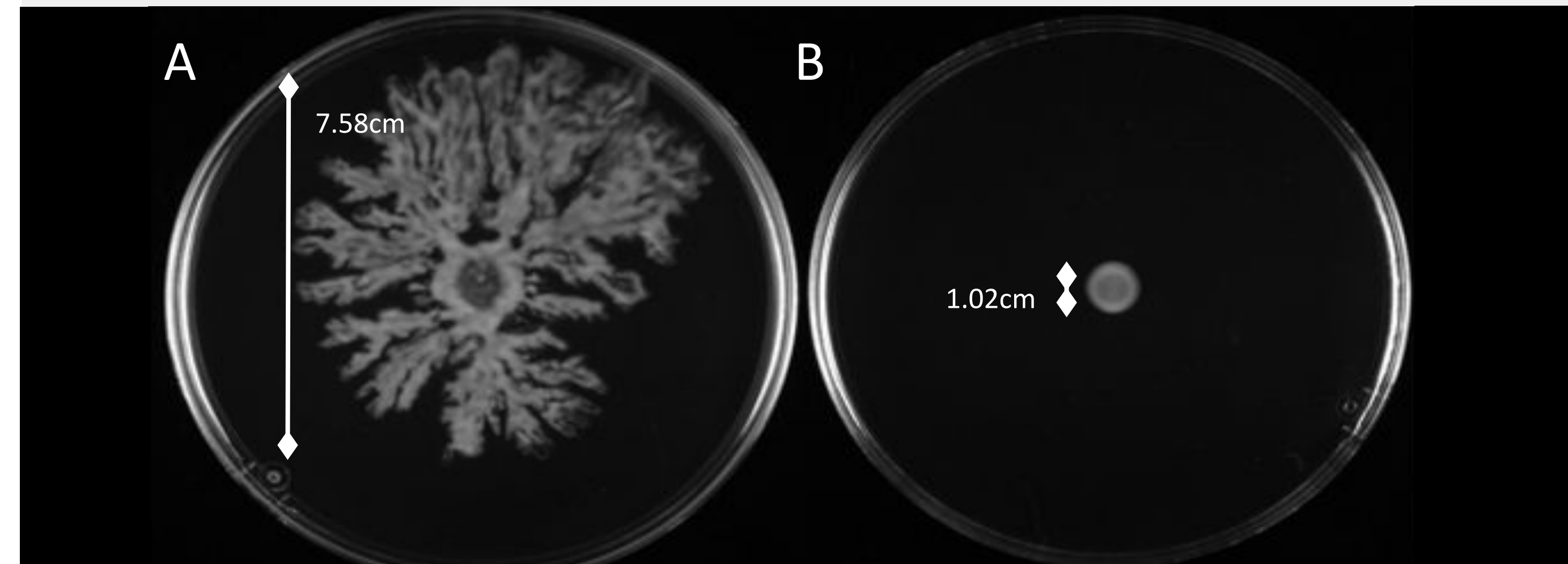


Figure II. *E. coli* CFT073 mutants growth on semi-solid swarming media [wild type (A) and $\Delta toI/C$ (B)].

We hypothesized that there may be a factor expressed within the wild-type supernatant that the $\Delta toI/C$ is unable to make. To test this, we grew both strains overnight then spun down and filtered their supernatants, before swapping them. The $\Delta toI/C$ mutant still did not exhibit any swarming (Fig III).

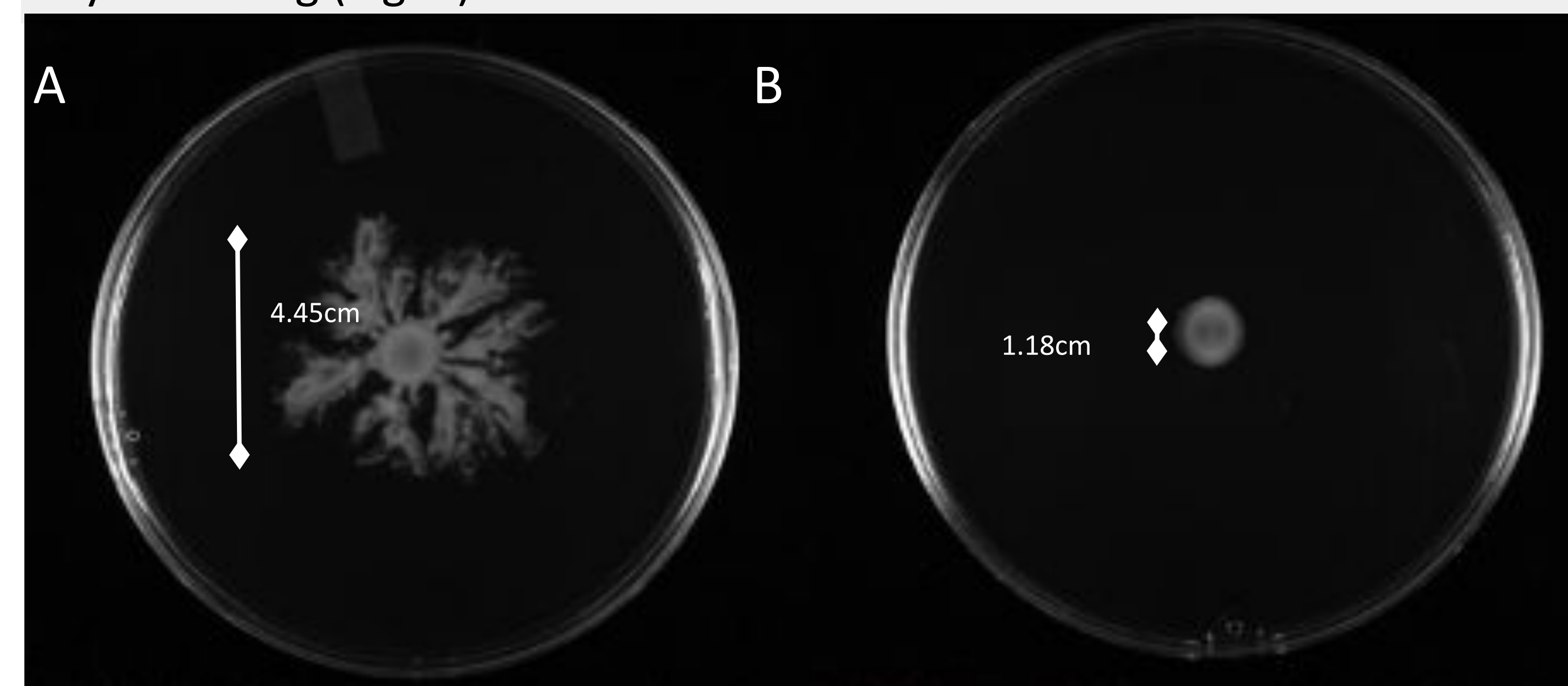


Figure III. *E. coli* CFT073 mutants growth on semi-solid swarming media with a liquid supernatant swap [wild type with $\Delta toI/C$ supernatant (A) and $\Delta toI/C$ with wild type supernatant (B)].

To test if a factor involved in swarming capability is only expressed when grown on solid media, we grew wild type strains on swarming media, and completed a supernatant switch using these cultures. The $\Delta toI/C$ mutant still did not exhibit any swarming (Fig IV).

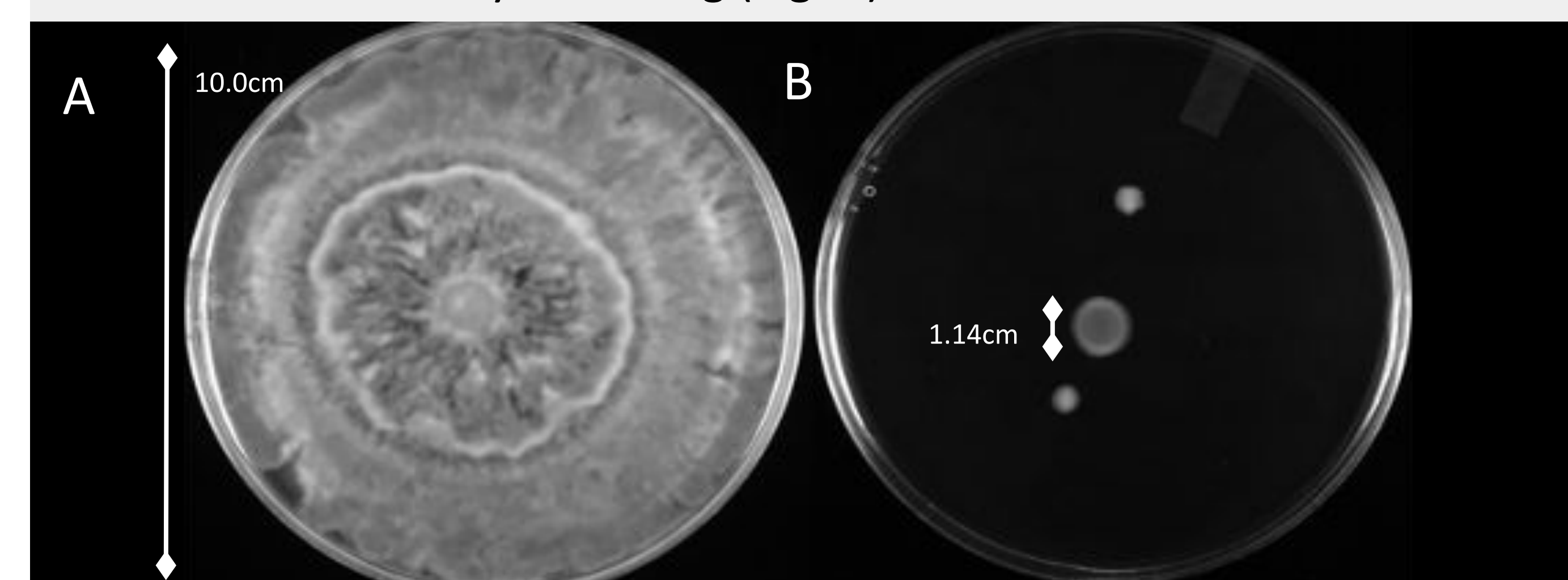


Figure IV. *E. coli* CFT073 mutants growth on semi-solid swarming media with a solid supernatant swap [wild type with $\Delta toI/C$ supernatant (A) and $\Delta toI/C$ with wild type supernatant (B)].

We hypothesized that one of the factors that wild type secretes which allows for swarming may be a surfactant. To test this, we added different concentrations of Triton X in an attempt to rescue the swarming phenotype in $\Delta toI/C$ (Fig V).

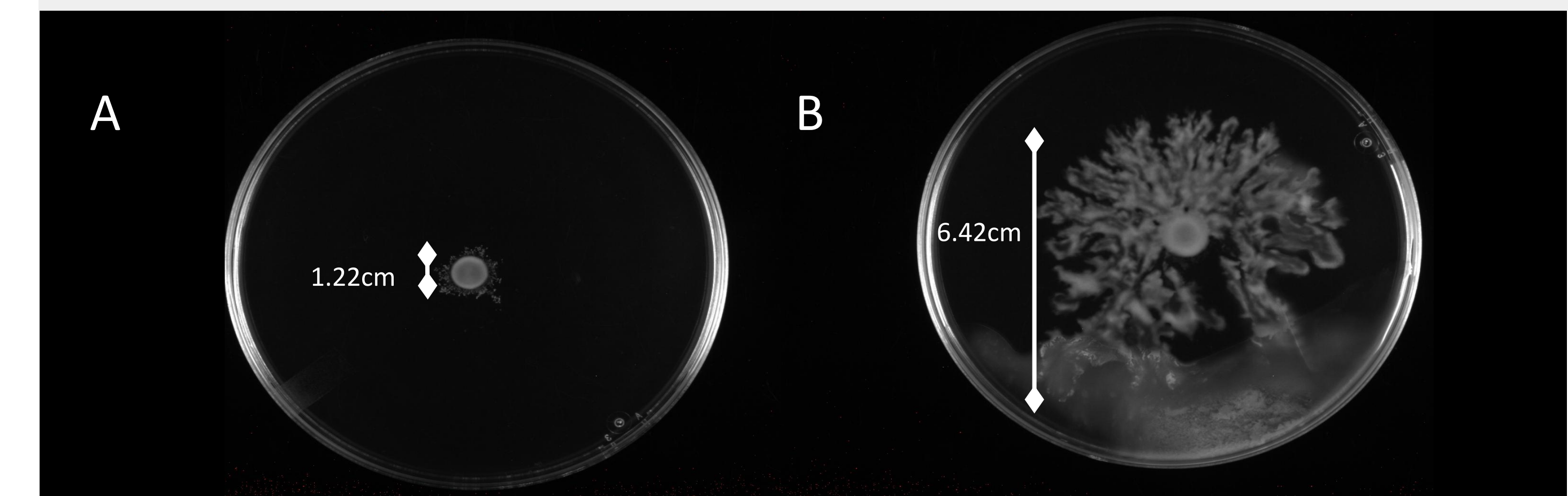


Figure V. *E. coli* CFT073 mutants growth on semi-solid swarming media with 0.0003% Triton X [$\Delta toI/C$ (A), wild type (B)].

Conclusions and Future Directions

Conclusions:

- After performing the original supernatant switch from liquid overnight cultures, and receiving no significant results, it was clear that the wild type supernatant as grown in TSB overnight was incapable of rescuing the swarming phenotype in the $\Delta toI/C$ mutant
- We performed another version of the supernatant switch in which the wild type was first grown on Eiken media and then collected in order to filter the supernatant and add to $\Delta toI/C$ cells
- Once again, the swarming phenotype was unable to be rescued, indicating that there may not be a component found in the supernatant that elicits swarming motility
- One hypothesis for what the wild type could be secreting is some sort of surfactant, a molecule that lowers surface tension
- To test this theory, we attempted to induce swarming by adding different concentrations of Triton X, a surfactant, to the media
- After testing various concentrations, we found that a 0.003% Triton X concentration rescues the swarming phenotype (Figure V)
- While this finding is significant as it informs us that surfactants have the capability of rescuing swarming, it is unclear whether this is the answer to why $\Delta toI/C$ cannot swarm

Future Directions:

- In the future, we hope to perform metabolomic analysis on the supernatants and test other theories such as secretion of quorum sensing molecules, in the hopes of identifying why $\Delta toI/C$ is unable to

References

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